Synthesizing the Voice of the Crowd: A Framework for Faithful Rating-Specific Opinion Summarization in Movie Reviews

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Abstract

The exponential growth of user-generated content has created a critical need for automated opinion summarization, yet a significant challenge remains in ensuring that AI-generated summaries faithfully reflect the specific sentiment intensity of a target rating group. This research addresses the "paradox of choice" in movie reviews by introducing a novel dual-phase framework designed to synthesize and validate "Representative Reviews" that maintain the authentic voice of specific rating cohorts.

First, to overcome the subjectivity of manual evaluation, we developed the "Judge" a fine-tuned DistilBERT model. Crucially, unlike standard classifiers, this model utilizes a **regression architecture** to measure sentiment intensity as a continuous value, allowing for precise quantification of the "sentiment distance" between AI-generated text and human ground truth. Second, using this objective metric, we conducted a comparative analysis of state-of-the-art Large Language Models **Gemini 2.5 Flash** and **Gemini 3.0 Flash**—across the entire rating spectrum. We evaluated the models' adaptability using two distinct prompting strategies: a direct "Basic Strategy" versus an emotional "Persona-Based Strategy".

Our findings reveal a distinct "U-shaped" performance curve: while Generative AI achieves near-perfect accuracy for extreme ratings (1 and 10 stars), it struggles significantly with mid-range nuances, often failing to distinguish a "good" movie (7 stars) from a "perfect" one. Furthermore, counter-intuitive results indicate that sophisticated **Persona-based prompting actually degraded performance**, increasing error rates compared to direct constraints. Crucially, we observed a divergence between semantic and sentiment fidelity: standard metrics (ROUGE, BERTScore) remained high despite significant rating errors, proving their insufficiency for detecting subtle emotional hallucinations. We conclude that current GenAI models function as "sentiment extremists," proficient at mimicking strong emotional polarities but lacking the calibration required to capture the subtlety of moderate human opinion.

1. Introduction

Online reviews have become the cornerstone of the modern entertainment economy. Platforms such as IMDb host millions of user-generated critiques that directly influence box office performance and audience consumption patterns [1]. However, this abundance of data has created a "paradox of choice" and significant information overload. For the average user, synthesizing a consensus from hundreds of disparate reviews is a cognitively demanding task [7]. While aggregate numerical ratings (e.g., "7.2/10") provide a quick metric, they fail to capture the qualitative reasons behind user satisfaction or frustration, such as specific

critiques of plot structure, acting nuances, or cinematographic choices [8].

Recent advancements in Generative AI and Large Language Models (LLMs) have enabled "Opinion Summarization"—the automatic generation of concise summaries from multiple source texts [3], [5]. Unlike standard text summarization, opinion summarization faces the unique challenge of sentiment preservation [6], [9]. A summary of 1-star reviews must not only convey the negative facts but also reflect the frustration, linguistic intensity, and specific vocabulary typical of that cohort. Conversely, a summary of 10-star reviews must capture the distinct patterns of appreciation and joy.

This research addresses the challenge of creating "Representative Reviews"—AI-generated summaries that maintain the specific "voice" of a rating group. We introduce a novel **dual-phase framework** designed to bridge the gap between generative capability and objective evaluation:

> **The "Judge" (Automated Sentiment Auditor):** Relying on manual evaluation for thousands of summaries is impractical and subjective. Therefore, we developed a robust, fine-tuned **DistilBERT-based evaluator** trained on human linguistic patterns. Crucially, unlike standard classifiers that simply categorize text, our "Judge" utilizes a **regression architecture** to measure sentiment intensity as a continuous value. This allows us to quantify the precise "sentiment distance" between the AI-generated summary and the ground truth (e.g., distinguishing a nuanced "7-star" review from an enthusiastic "9-star" one).
>
> **The Summarizer (Generative Experimentation):** We conduct a comparative analysis of state-of-the-art Large Language Models—specifically **Gemini 2.5 Flash** and **Gemini 3.0 Flash**—to synthesize faithful reviews. To test the models' adaptability, we employ distinct prompting strategies [4]: a **Basic Strategy** (direct constraints) versus a **Persona-Based Strategy** (emotional priming, e.g., "act as an angry critic"). This comparison aims to identify which configuration best replicates the authentic "voice" of the crowd across the entire rating spectrum [2, 10].

2. Literature Review

This section reviews existing research on movie review analysis, the evolution of automatic summarization techniques, and the specific challenges of maintaining sentiment consistency in GenAI outputs.

2.1 Linguistic Patterns in Movie Reviews

Understanding the baseline behavior of human reviewers is essential for any summarization task. Previous research on IMDb reviews has demonstrated that linguistic patterns vary significantly across rating levels. Ren and Chandra [1] analyzed approximately 138,000 reviews and found that extreme ratings (1 and 10 stars) are characterized by high emotional intensity and distinct vocabulary, whereas moderate ratings (4-8 stars) tend to be longer and more analytical. Furthermore, their findings indicate a negative correlation between rating and abusive language, with lower-rated reviews containing significantly more offensive content [1]. These findings serve as the "ground truth" for our project: a valid "Representative Review" must replicate these specific linguistic signatures to be considered authentic.

2.2 From Extractive to Abstractive Opinion Summarization

Opinion summarization is a sub-field of Multi-Document Summarization (MDS) that focuses on extracting or abstracting subjective information from multiple sources [3]. Early approaches were predominantly extractive, selecting key sentences directly from the source text to form a summary. For instance, Angelidis et al. [9] proposed the "Quantized Transformer" (QT), an unsupervised model that clusters similar opinions to generate summaries. While effective at extracting facts, these methods often result in disjointed text that lacks the flow of a natural review. The emergence of Large Language Models (LLMs) like GPT-4 has shifted the focus to abstractive summarization, where the model generates novel text to synthesize information [2]. Recent studies have shown that LLMs can effectively cluster and summarize user reviews by identifying common themes (e.g., "battery life" in products or "acting" in movies) [4]. Simmering and Huoviala [8] demonstrated that LLMs outperform traditional models in aspect-based sentiment analysis, allowing for more granular control over the summary's content.

2.3 Challenges in Controllable and Faithful Summarization

Despite the capabilities of LLMs, significant challenges remain in generating faithful summaries. A major issue is hallucination, where the model invents details not present in the source reviews. Kang et al. [10] highlight that abstractive models often generate plausible but factually incorrect content due to domain bias, necessitating new decoding strategies to mitigate this risk. Another critical challenge is sentiment preservation. Wallace et al. [7] found that many multi-document summarization models fail to accurately reflect the "average critic consensus," often becoming over-sensitive to input ordering or under-sensitive to the ratio of positive-to-negative reviews. This aligns with the findings of Mousiadis et al. [6], who argue that standard models struggle with controllable summarization when the context (such as a specific rating or sentiment target) is complex. Techniques like "Review Stratification" have been proposed to group reviews by score before summarization to prevent positive and negative sentiments from canceling each other out [5].

2.4 Proposed Contribution

Our project bridges these gaps by focusing specifically on Rating-Specific Abstractive Summarization. We propose a framework that not only uses GenAI to summarize clusters of same-rated reviews but also introduces a rigorous validation mechanism. Unlike previous works that rely solely on human evaluation or text overlap metrics like ROUGE which often fail to capture sentiment accuracy [3, 5], we utilize a fine-tuned BERT classifier trained on human linguistic patterns identified by Ren and Chandra [1]. This allows us to quantitatively measure whether the AI-generated summary successfully mimics the "voice" of a real human reviewer at that specific rating level, addressing the call for better evaluation metrics in the field.

3. Methodology

To rigorously answer our research question, we designed a multi-stage experimental framework combining supervised learning (for evaluation) and generative AI (for content creation).

3.1 Data Preparation

We utilized the "IMDb Movie Reviews with Ratings" dataset [11], a widely recognized collection comprising 50,000 user-generated critiques. Sourced from Kaggle (Chodavadiya) and derived from the foundational Large Movie Review Dataset (Maas et al., 2011), this corpus serves as a benchmark for both

binary and fine-grained sentiment analysis.

Class Structure and "Polarity Gap":

A defining characteristic of this dataset is its intentional exclusion of "neutral" opinions to ensure clear sentiment distinction. Specifically, the original curators omitted any reviews rated 5 or 6 stars. Consequently, the data is not continuous but rather segmented into **8 distinct ordinal classes**: Negative ratings ($R \in \{1,2,3,4\}$) and Positive ratings ($R \in \{7,8,9,10\}$). This structure presents a unique challenge for the model, forcing it to distinguish between subtle gradations of "good" and "bad" (e.g., distinguishing a 3 from a 4) without the buffer of neutral examples.

Distribution and Balance:

At the binary level, the dataset is perfectly balanced, containing exactly 25,000 positive and 25,000 negative reviews. However, the distribution *within* these polarities exhibits the non-uniform "J-shaped" curve often observed in online feedback, where extreme ratings (1 and 10) are typically more frequent and linguistically distinct than moderate ones (4 or 7). Addressing this intra-class imbalance was critical, as our "Judge" model required robustness to identify the nuanced linguistic markers of the under-represented intermediate ratings, preventing it from overfitting to the high-intensity vocabulary of the extremes.

Data Splitting:

To simulate a rigorous evaluation environment, we partitioned the dataset into two subsets: an **80% Training Set** (40,000 reviews) used to fine-tune the BERT-based "Judge" evaluator, and a held-out **20% Test Set** (10,000 reviews) reserved for the generative experiments and final validation.

3.2 Phase 1: The Evaluator Model ("The Judge")

Before generating synthetic reviews, it was essential to establish an objective, scalable metric for assessing sentiment fidelity. Since manual evaluation of thousands of machine-generated outputs is impractical and prone to subjective bias, we developed an automated "Judge." We fine-tuned a **DistilBERT** model to predict the star rating of a given text, effectively serving as a ground-truth proxy.

We explored two distinct architectural approaches for this task:

1. **Multi-Class Classification:** Treating each rating as a distinct, independent category.
2. **Regression:** Treating the rating as a continuous numerical value to capture sentiment proximity.

The following sections detail the training process and performance metrics of these architectures.

3.2.1 Training the Multi-Class Classifier (Baseline)

Initially, we modeled the problem as a standard multi-class classification task. The dataset was filtered to include only the target rating groups $R \in \{1,2,3,4,7,8,9,10\}$, which were mapped to distinct categorical labels (0-7). The dataset was split into training (80%) and validation (20%) sets.

Model Configuration:

We utilized DistilBertForSequenceClassification with a classification head configured for 8 output labels. The model was trained using Cross-Entropy Loss, which penalizes any deviation from the exact class label equally.

Results and Analysis:

The classification model demonstrated strong performance in identifying distinct sentiment polarities but struggled with granular ordinal differences.

> **High Polarity Accuracy:** As evidenced by the confusion matrix, the model achieved high precision for extreme ratings. For example, **Rating 1** was correctly identified in 1,143 instances (out of ~1,300), and **Rating 10** showed similarly high recognition rates.
>
> **The "Ordinality" Problem:** A significant limitation was observed in the intermediate ratings. The model frequently confused adjacent classes, such as misclassifying **Rating 4** as **Rating 3** (147 instances) or **Rating 8** as **Rating 9**.
>
> **Metric Limitation:** In a classification framework, predicting a "3" when the ground truth is "4" is penalized as a complete error (Accuracy = 0), equivalent to predicting a "1". However, semantically, a "3" is a much better prediction than a "1". This "all-or-nothing" penalty obscured the model's ability to understand sentiment proximity, leading to the conclusion that a regression-based approach (Section 3.2.2) would be more suitable for evaluating the subtle "Voice of the Crowd."

3.2.2 Training the Regression Model (The Final Judge)

To address the limitations of the classifier, we re-architected the Judge as a **Regression Model**. We modified the DistilBERT architecture to output a single scalar value (num_labels=1) and utilized **Mean Squared Error (MSE)** as the loss function.

Performance:

The regression approach significantly improved the system's ability to gauge sentiment distance.

> **Granularity:** Instead of discrete errors, the model outputs continuous predictions (e.g., predicting 7.8 for a target of 8), allowing for a more nuanced evaluation of the Generative AI outputs.
>
> **Error Minimization:** By minimizing the numerical distance between the predicted and actual rating, the regression model penalized "near misses" (e.g., 3 vs. 4) far less than "catastrophic misses" (e.g., 1 vs. 10), aligning the automated metric with human intuition regarding sentiment similarity.

3.2.3 Optimization

To further combat overfitting, we implemented:

Extended Context: Increased max_length from 256 to 512 tokens.

Regularization: High dropout rates (30%) in the DistilBERT configuration.

Hyperparameters: Learning rate of 2e-5, weight decay of 0.3, and Early Stopping with a patience of 4 epochs.

3.3 Experimental Procedure & Pipeline Architecture

To empirically evaluate the research question, we developed an automated data processing pipeline designed to simulate mass review generation and iterative auditing. The experiment was conducted across all target rating groups $R \in \{1,2,3,4,7,8,9,10\}$ using the following systematic protocol:

3.3.1 Cluster Sampling (Data Retrieval)

In each iteration, the pipeline retrieves a random sample of ten user reviews ($N = 10$) sharing an identical star rating from the IMDb dataset. This cluster represents a homogeneous "voice of the crowd" for that specific sentiment level. Random sampling with replacement was employed to ensure diversity across the 100 experimental runs per rating group.

3.3.2 Generative Prompt Strategies

The sampled cluster is fed into Google's state-of-the-art Large Language Models using the official developer API. To ensure precise model versioning, we generated secure API keys via Google AI Studio and targeted two specific endpoints: Gemini 2.5 Flash and Gemini 3.0 Flash.

Following this integration, we employed two distinct prompt engineering strategies to test different aspects of sentiment synthesis. **The exact specifications for both prompt templates are detailed in Listing 1 below.**

> **Basic Prompt (Baseline):** A direct instruction to the model to read the provided reviews and synthesize a single "Representative Review." The model is explicitly constrained to aggregate the collective sentiment and writing style without adding external bias or meta-text.
>
> **Persona-Based Prompt:** A dynamic prompting approach where the model is instructed to act as a "typical IMDb user" rather than a neutral summarizer. Crucially, instead of assigning a static emotion, the prompt utilizes **adaptive mimicry**: the model is explicitly directed to analyze the provided reviews and adopt their specific "tone, sentiment intensity, and vocabulary" (e.g., mimicking slang-heavy frustration or analytical praise). This tests the hypothesis that instructing the model to "act" like the source authors yields more authentic sentiment representation than standard summarization.

Strategy	Prompt Template
Basic Strategy	"Please analyze the following collection of user reviews for a movie, all of which gave a rating of {RATING}. Your task is to generate a single 'Representative Review' that summarizes the consensus opinions found in these texts. The summary should cover the main pros and cons mentioned by the users but must be written as if it is a single, coherent review written by one person.Input Reviews: {INPUT_REVIEWS}"
Persona Strategy	"Act as a typical IMDb user who has just watched this movie and decided to give it a rating of {RATING}. I will provide you with several actual reviews from other users who gave the same rating. Your goal is to write a new review that captures the 'voice of the crowd.' Requirements: Adopt the specific tone, sentiment intensity, and vocabulary used in the provided reviews (e.g., if they are analytical, be analytical; if they are angry/slang-heavy, use similar language). Do not sound like an AI summarizer. Sound like a human movie fan. Reflect the specific reasons for satisfaction or frustration found in the text. Input Reviews: {INPUT_REVIEWS}"

Listing 1: Exact Prompt Specifications Used in Experiments

3.3.3 The Evaluation Loop and Metrics

The synthetic review generated by the LLM is immediately passed as input to our fine-tuned DistilBERT regression model (the "Judge"). The Judge predicts a floating-point sentiment score, which is then mapped to the nearest valid integer rating.

To mitigate the stochastic nature of LLMs, this process is repeated 100 times for each rating group. Performance is evaluated using two primary metrics:

1. **Exact Match Accuracy:** The percentage of runs where the Judge's predicted rating exactly matches the target rating.
2. **Mean Absolute Error (MAE):** The average absolute difference between the target rating (y) and the predicted rating , providing a granular measure of sentiment deviation.

4. Results and Analysis

4.0.1 "Judge" Model Performance

The regression-based Judge model demonstrated a significant improvement in logical consistency over the classification baseline. As illustrated in Figure 1, the model reached its optimal performance at Epoch 17, achieving a Validation Loss of 2.518 and a Mean Absolute Error (MAE) of approximately 1.00. On average,

the model's prediction deviates by only 1 star from the true rating.

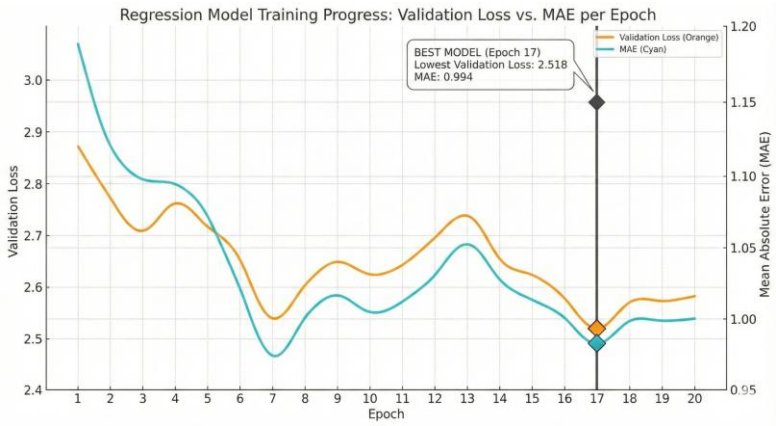

Figure 1: Training progress of the Regression Judge model, showing the convergence of Validation Loss and Mean Absolute Error (MAE) over 20 epochs.

While the exact accuracy remained similar to the classification attempt, the **Confusion Matrix** revealed a crucial improvement: errors were tightly clustered around the diagonal. For instance, when the true label was 10, the model predicted 10 (1319 times), 9 (388 times), and 8 (147 times), but rarely predicted 1. This confirms the model successfully learned the sentiment spectrum.

To answer our research question regarding which generative AI model most accurately reflects requested movie ratings, we conducted a comparative analysis of **Gemini 2.5 Flash** and **Gemini 3.0 Flash**. We generated reviews for specific target ratings (1-4 and 7-10) using two distinct prompt strategies ("Basic" and "Persona") and evaluated the results using a classifier trained on human IMDb reviews.

The analysis focuses on three key metrics: **Mean Absolute Error (MAE)** (the distance between the requested rating and the predicted rating) and **Accuracy Percentage** (how often the model hit the exact specific rating).

4.1 Overall Performance by Target Rating

As shown in Figure 2, the data reveals a distinct "U-shaped" performance curve for both models. Both models demonstrated near-perfect accuracy at the extreme ends of the rating spectrum but struggled significantly with mid-range nuances.

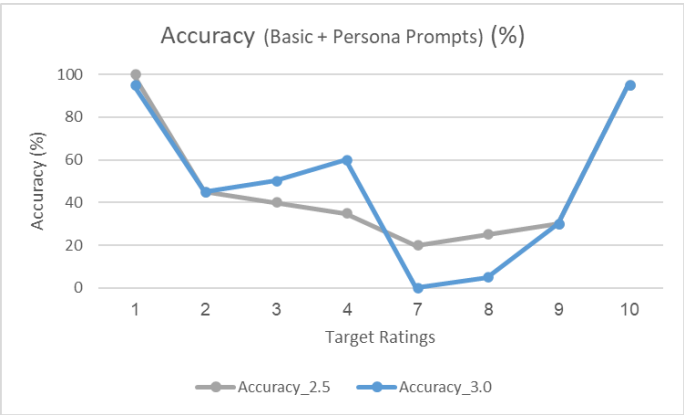

Figure 2: Comparative accuracy percentages for Gemini 2.5 and 3.0 across specific target ratings, illustrating the "U-shaped" performance curve.

Table 1: Comparative Performance per Target Rating (Aggregated)

Table 1 presents a detailed breakdown of the performance metrics per target rating. While both models perfectly capture "hatred" (Rating 1) and "perfection" (Rating 10), significant divergences appear in the intermediate ratings, particularly at Rating 4 where Gemini 3.0 shows improved nuance.

Target Rating	Mean Error (2.5)	Mean Error (3.0)	Accuracy % (2.5)	Accuracy % (3.0)	Analysis
1	**0.00**	0.05	**100%**	95%	Both models perfectly capture "hatred."
2	0.55	0.55	45%	45%	Moderate difficulty distinguishing 2 from 1 or 3.
3	1.05	**0.60**	40%	**50%**	Gemini 3.0 shows improved nuance here.
4	1.40	**0.85**	35%	**60%**	**Significant Win for 3.0.** It better understands "mediocre/bad."
7	**1.75**	1.85	**20%**	0%	**Major Failure Point.** Both models struggle to define a "7".
8	**1.10**	1.40	**25%**	5%	Gemini 2.5 outperforms 3.0 in this

					range.
9	0.70	0.70	30%	30%	Both models often overshoot towards 10.
10	0.05	0.05	95%	95%	Both models perfectly capture "perfection."

Table 1: Detailed performance breakdown per rating, comparing Mean Error and Accuracy between Gemini 2.5 and Gemini 3.0.

Key Findings:

Key findings regarding error distribution are highlighted in Figure 3. Extreme Polarity Mastery: For ratings 1 and 10, both models achieved ≈ 100% accuracy with near-zero error. This suggests that Generative AI easily mimics the linguistic markers of extreme emotions (anger/frustration for 1, joy/adoration for 10), which aligns with the GloVe/DistilBERT findings in the related study regarding distinct vocabulary for high vs. low scores.

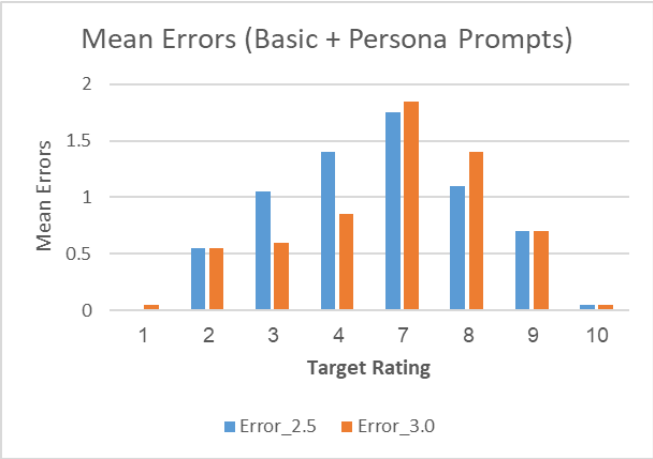

Figure 3: Mean Absolute Error (MAE) distribution by target rating, highlighting the difficulty in synthesizing mid-range sentiment.

The "Mid-Range" Confusion: The highest error rates occurred at ratings **4** and **7**.

Rating 7 was the hardest category for both models. Gemini 3.0 failed to produce a single accurate "7" rating (0% accuracy), often producing reviews that the classifier interpreted as an 8 or 9.
Rating 4 showed the biggest divergence. Gemini 2.5 struggled (1.4 Error), while Gemini

3.0 significantly improved performance (0.85 Error), suggesting 3.0 has a better semantic grasp of "below average but not terrible."

4.2 Impact of Prompt Engineering: Basic vs. Persona

A counter-intuitive finding of this research, depicted in Figure 4, is that sophisticated prompting ("Persona") actually reduced the accuracy of the generated reviews compared to "Basic" prompting.

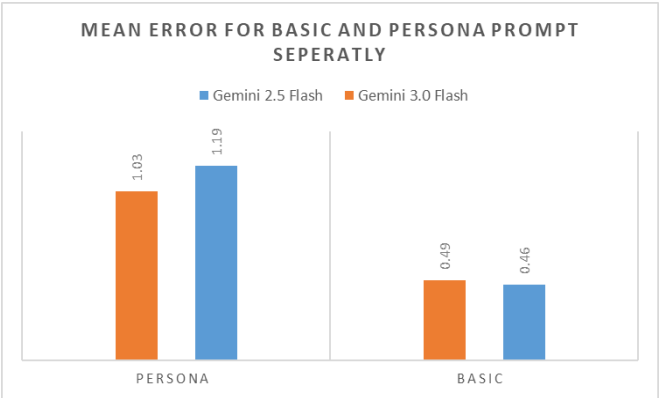

Figure 4: The impact of prompt strategies on error rates, demonstrating higher Mean Absolute Error (MAE) for Persona-based prompting across both models.

Table 2 quantifies this discrepancy. The analysis shows that the Basic prompt is approximately 61% more accurate for Gemini 2.5 and 52% more accurate for Gemini 3.0. This suggests that for strict classification tasks, concise instructions produce output that aligns better with the classifier's training data.

Model	Basic Prompt Error	Persona Prompt Error	Conclusion
Gemini 2.5 Flash	**0.46**	1.19	Basic is approximately 61% more accurate.
Gemini 3.0 Flash	**0.49**	1.03	Basic is approximately 52% more accurate.

Table 2: Aggregated Mean Error comparison between Basic and Persona prompt strategies.

Analysis of the Persona Failure:

When instructed to adopt a persona (likely a critic or cinephile), the models likely introduced complex vocabulary, nuanced sentence structures, or "fluff" text that confused the classifier.

For **Gemini 2.5**, the Persona prompt caused the error rate for Rating 7 to spike to **2.5**, meaning when asked for a 7, it often wrote reviews that sounded like a 4 or a 10.

The **Basic prompt** yielded the most consistent results, suggesting that for strict classification tasks, concise and direct instructions produce output that aligns better with the training data of standard classifiers.

4.3 Model Comparison: Is Newer Better?

Comparing Gemini 2.5 Flash to the newer Gemini 3.0 Flash yields a mixed verdict rather than a clear victory for the newer model.

1. **Negative Nuance (Ratings 1-4): Gemini 3.0 is superior.** It significantly reduced the error for Rating 3 (0.6 vs 1.05) and Rating 4 (0.85 vs 1.4). It appears 3.0 is better at writing reviews that express "disappointment" without resorting to "hatred."
2. **Positive Nuance (Ratings 7-10): Gemini 2.5 is superior.** For Rating 7, Gemini 2.5 had a lower error (1.75 vs 1.85) and managed 20% accuracy where 3.0 had 0%. Gemini 3.0 appears to suffer from "positivity bias" in this range, making a 7 or 8 sound too much like a perfect 10.

4.4 Textual Similarity vs. Sentiment Fidelity

To complement our sentiment-based evaluation, we calculated standard text-overlap metrics (ROUGE) and semantic similarity metrics (BERTScore) for all model configurations. The results, summarized in Figure 5, reveal a significant divergence between traditional metrics and sentiment accuracy.

Low ROUGE, High Semantic Consistency:

Across all configurations, ROUGE scores were consistently low (ranging from 0.076 to 0.090), while BERTScores remained high (>0.81). This confirms that the models are performing abstractive summarization generating novel sentences rather than extracting phrases from the source text. The high BERTScore indicates that while the specific wording differs from the source, the semantic content remains relevant.

Model Comparison:

Interestingly, Gemini 3.0 Flash achieved slightly higher ROUGE scores (0.090) compared to Gemini 2.5 (0.076-0.079) in both prompting strategies. This suggests that the newer model may adhere somewhat more closely to the structural patterns of the source reviews.

The Metric Discrepancy:

Crucially, these metrics remained relatively stable across "Basic" and "Persona" prompts, failing to capture the dramatic drop in sentiment accuracy observed in Section 4.2. This validates our hypothesis that standard linguistic metrics like ROUGE and BERTScore are insufficient for measuring emotional fidelity, further necessitating the use of a specialized sentiment regression model (the "Judge").

4.5 Conclusion of Analysis

The data suggests that while Generative AI is highly effective at mimicking clear sentiment (hate/love), it struggles with the calibration of moderate opinions. The "related study" noted that rewatchability and complex emotions appear across ratings; our results indicate that the AI struggles to inject these subtleties into a "7" or "4" rating without tipping the scale too far in either direction.

Interestingly, Gemini 2.5 Flash (Basic Prompt) remains the most reliable overall performer for positive ratings, while Gemini 3.0 Flash (Basic Prompt) is the superior choice for nuanced negative ratings. Figure 5 provides a comprehensive summary of these behavioral phenotypes, detailing the Accuracy and key confusion clusters for each model configuration.

Generative Model	Prompt Strategy	Accuracy (%)	Mean Absolute Error (MAE)	ROUGE Score	BERTScore	Behavioral Phenotype	Key Confusion Clusters
Gemini 2.5 Flash	Basic	60.0%	0.46	0.076	0.812	**Positive Nuance Capable:** Better at distinguishing 'good' (7-8) from 'perfect' (10), but struggles with negative nuance.	4→3 (Moderate), 7→9 (High Error)
Gemini 3.0 Flash	Basic	53.75%	0.49	0.090	0.815	**Negative Nuance Specialist:** Excellent at distinguishing 'bad' (4) from 'terrible' (1), but fails completely at 'good' ratings.	✗→9 (Severe Failure - 0% Acc), ↓ 8→9
Gemini 2.5 Flash	Persona-Based	37.5%	1.19	0.079	0.813	**Complexity Overload**: The persona adds noise that confuses the classifier, causing severe drift in mid-range ratings.	7→10 (Extreme Drift), 3→1
Gemini 3.0 Flash	Persona-Based	41.25%	1.03	0.090	0.812	**Instability**: Similar to 2.5, the persona degrades performance, though slightly more stable than 2.5 in error magnitude.	7→10 (Extreme Drift), 4→2

Figure 5: Comprehensive summary of model phenotypes, confusing clusters, and scores across all experimental configurations.
*** MAE - Lower means better performance**

5. Discussion

This study aimed to evaluate whether Generative AI models (Gemini 2.5 Flash and Gemini 3.0 Flash) could generate movie reviews that accurately reflect specific numerical ratings (1-4 and 7-10) when measured against a classifier trained on human behavior. By integrating findings from previous research on human review patterns with our experimental data, several key insights emerge.

5.1. The "Polarity Trap" and Human Linguistic Patterns Our results demonstrate a striking dichotomy: both AI models achieved near-perfect accuracy (95-100%) for extreme ratings (1 and 10) but failed significantly in the mid-range (4 and 7). This aligns with the findings of the related study, which noted that high-rated reviews are characterized by distinct positive expressions (joy, appreciation) and low-rated reviews by offensive language and anger. It appears that Generative AI easily mimics these distinct "emotional extremes." However, the models struggle with the subtlety required for intermediate ratings. When asked for a "7" (a good but not perfect movie), the models likely overused the positive vocabulary

associated with "10," causing the classifier to misinterpret the review as a masterpiece rather than a solid film.

5.2. The Counter-Intuitive Role of Persona A surprising finding was that the "Persona" prompt (acting as a movie critic) consistently performed worse than the "Basic" prompt. While one might expect a persona to add depth, it likely introduced complex sentence structures or sophisticated vocabulary that deviated from the "average user" language the classifier was trained on. This suggests that for tasks requiring mimicry of general human data (like IMDb reviews), simple prompting is more effective than elaborate role-playing.

5.3. Model Evolution: Nuance vs. Enthusiasm Comparing the two models reveals a shift in behavior. Gemini 3.0 demonstrated a superior ability to handle negative nuance (Ratings 3-4), significantly outperforming Gemini 2.5 in distinguishing "mediocre" from "terrible". Conversely, Gemini 2.5 remained better at positive nuance. This suggests that while newer models may have improved semantic reasoning for criticism, they may still suffer from a "positivity bias," struggling to write a positive review that isn't overwhelmingly enthusiastic.

5.4. The Dissonance Between Semantic and Sentiment Metrics A critical insight from this study is the insufficiency of traditional summarization metrics for sentiment-sensitive tasks. As detailed in Section 4.4, both models maintained high BERTScores (>0.81) across all ratings, implying that the generated reviews were semantically coherent and relevant to the source material. However, this high semantic consistency masked the significant "sentiment hallucination" revealed by our specialized Judge model.

While standard metrics suggested the models were performing well (generating plausible movie reviews), our regression-based MAE metric exposed that the *intensity* of the opinion was frequently misaligned, particularly in mid-range ratings. This confirms that general-purpose metrics like ROUGE or BERTScore cannot detect when a "7-star" review drift into "10-star" enthusiasm. Consequently, reliance on specific sentiment auditors (like our fine-tuned DistilBERT) is essential for validating Generative AI outputs in subjective domains.

6. Conclusion

This project set out to determine if Generative AI can serve as a faithful proxy for human movie reviewers across the rating spectrum. Using a dataset of 50K IMDb reviews and a comparative analysis of Google's Gemini models, we arrive at the following conclusions:

1. **AI is an "Extremist":** Generative AI is currently highly effective at generating content that reflects strong sentiment (absolute hate or absolute love) but lacks the calibration to accurately reflect moderate opinions. It struggles to generate the "gray area" of human opinion—movies that are just "okay" or "pretty good."
2. **Newer is Not Always Better for All Tasks:** While Gemini 3.0 Flash showed improvements in generating nuanced negative reviews, it did not universally outperform the older Gemini 2.5 Flash, particularly in the positive rating spectrum.
3. **Inadequacy of Standard Metrics:** Our results prove that traditional text similarity metrics

(ROUGE, BERTScore) are insufficient for evaluating opinion summarization. While the AI successfully mimicked the *structure* and *semantics* of reviews (high BERTScore), it failed to capture the accurate *emotional calibration* (low Judge accuracy). This validates the necessity of our "Judge" framework, demonstrating that AI can produce linguistically fluent text that is nonetheless "sentimentally hallucinatory."

Future Work:

Future research should focus on "sentiment calibration" techniques. Rather than zero-shot prompting, fine-tuning models on specific rating datasets (e.g., feeding the model examples of "7-star" reviews) could help bridge the gap between AI generation and human subtlety. Additionally, investigating why "Persona" prompting degrades performance could lead to better guidelines for prompt engineering in data augmentation tasks.

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